1 Trigonometric Equations and Applications

Central question. When a trigonometric function has a prescribed value, how can we find *all* the angles that produce it? How can we use angles to solve triangles that are not right triangles?

1.1 A first equation on the unit circle

Consider $\sin x = 1/2$. The unit circle has two points whose vertical coordinate is $1/2$, at $x = \pi/6$ and $x = 5\pi/6$. Every full revolution reproduces these points:

$$\boxed{x = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad x = \frac{5\pi}{6} + 2\pi k, \qquad k \in \mathbb{Z}.}$$

An inverse trigonometric function gives one principal angle, not necessarily every solution. The unit circle and periodicity give the complete answer.

Example 1.1. Solve $\cos x = -\sqrt{2}/2$ for $0 \le x < 2\pi$. The solutions are $3\pi/4$ and $5\pi/4$.

1.2 Equations with transformations

To solve $\sin(2x) = \sqrt{3}/2$ on $[0, 2\pi)$, first solve for $u = 2x$ on $[0, 4\pi)$. We obtain $u = \pi/3, 2\pi/3, 7\pi/3, 8\pi/3$ and therefore

$$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}.$$

A changed frequency changes how many full cycles fit into the requested interval.

Example 1.2. Solve $2\cos x - 1 = 0$ on $[0, 2\pi)$. Isolating cosine gives $\cos x = 1/2$, hence $x = \pi/3$ or $5\pi/3$.

1.3 Factoring and checking domains

An equation may contain more than one trig function. Algebraic techniques still work:

$$2\sin^2 x - \sin x = 0 \quad \Longrightarrow \quad \sin x(2\sin x - 1) = 0.$$

On $[0, 2\pi)$ we obtain $x = 0, \pi, \pi/6, 5\pi/6$. Dividing by $\sin x$ at the outset would have lost two solutions.

1.4 Trigonometric identities in equations

For example, $2\cos^2 x - 1 = 0$ becomes $\cos^2 x = 1/2$, hence $\cos x = \pm\sqrt{2}/2$. There are four solutions on $[0, 2\pi)$. Using identities can make equations simpler, but each transformation should preserve the original domain.

1.5 Solving triangles: notation

Write the vertices of a triangle as A, B, C, and the opposite side lengths as a, b, c. Its angles satisfy $A + B + C = \pi$. For right triangles, the six trig ratios are enough. For oblique triangles, two laws extend our tools.

Definition 1.1 (Law of Sines)**.** For every nondegenerate triangle,

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.}$$

Example 1.3. A triangle has $A = 40°$, $B = 65°$, and $a = 9$ cm. Since $C = 75°$, the Law of Sines gives $b = 9\sin 65°/\sin 40° \approx 12.69$ cm.

1.6 The ambiguous SSA case

Given two sides and a nonincluded angle (SSA), the Law of Sines may produce zero, one, or two triangles. If $\sin B = s$, both $B = \arcsin s$ and $B = \pi - \arcsin s$ must be considered, and each candidate must fit with the known angle A in a triangle.

Example 1.4. Suppose $A = 30^\circ$, $a = 5$ and $b = 8$. Then $\sin B = 8 \sin 30^\circ / 5 = 0.8$. Both $B \approx 53.13^\circ$ and $B \approx 126.87^\circ$ are consistent with $A + B < 180^\circ$, so two distinct triangles exist.

Definition 1.2 (Law of Cosines)**.** For a triangle with sides a, b, c opposite angles A, B, C,

$$\boxed{c^2 = a^2 + b^2 - 2ab\cos C.}$$

When $C = 90^\circ$ this becomes the Pythagorean theorem. The cosine term measures how the included angle modifies the length of the opposite side.

Example 1.5. Two sides of a triangle are 7 m and 10 m, with included angle 60°. The opposite side satisfies $c^2 = 49 + 100 - 140(1/2) = 79$, so $c = \sqrt{79}$ m.

1.7 Applied modeling and interpretation

Two sensors observe a distant object from different locations. The baseline and their sight angles make a triangle. The Law of Sines can recover an inaccessible distance if we know a side and two angles. Always include the diagram, angle units, and a reasonableness check.

1.8 Exercises

1. Solve $\sin x = -\sqrt{3}/2$ for $0 \le x < 2\pi$.
2. Solve $2\sin(3x) = 1$ for $0 \le x < 2\pi$; find every solution.
3. Solve $2\cos^2 x - \cos x = 0$ on $[0, 2\pi)$.
4. Solve $\tan x = -1$ for all real x.
5. A triangle has $A = 35^\circ$, $B = 80^\circ$ and $a = 12$. Find C and b.
6. For $A = 30^\circ$, $a = 5$, $b = 8$, calculate both possible values of C in the SSA case.
7. Find the third side of a triangle when two sides have length 6 and 9 and their included angle is 120°.
8. Explain why the Law of Cosines becomes the Pythagorean theorem for right triangles.